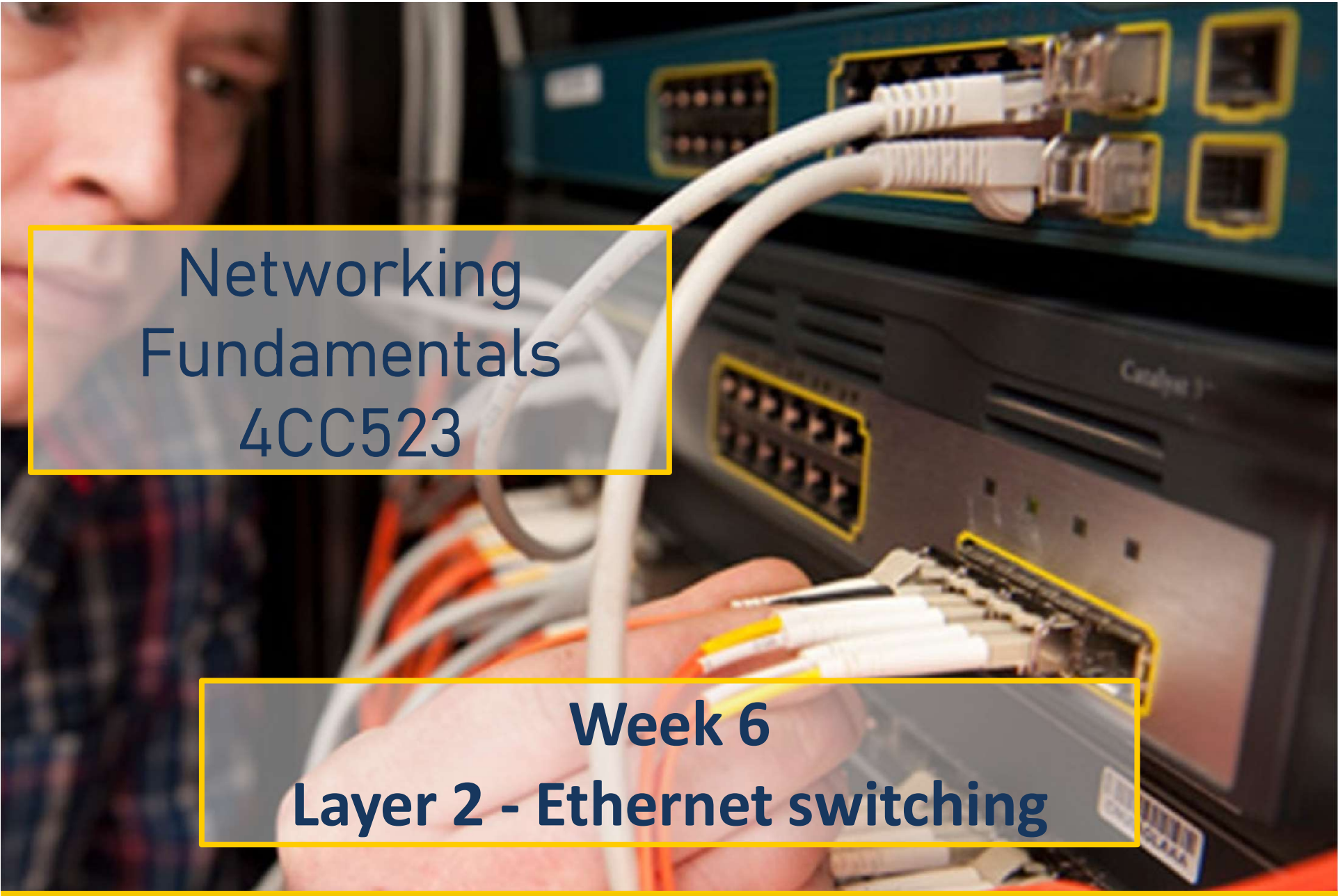


A close-up photograph of a person's face, partially visible on the left, looking towards the camera. In the background, there is a network switch or router with several Ethernet cables plugged into its ports. The cables are white and orange. The person's hand is visible, holding one of the cables. The text "Networking Fundamentals 4CC523" is overlaid on the image in a blue font, enclosed in a yellow rectangular box.

Networking Fundamentals 4CC523

Week 6

Layer 2 - Ethernet switching



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DERBY

Individual Health & Safety Responsibilities

To help prevent the spread of the COVID-19 virus we ask you to carry out the following, to help keep everyone safe & aid with contact tracing.

- If you have any COVID-19 symptoms leave the class immediately if possible & return to your accommodation. Contact the NHS on 111 & then contact student wellbeing.
- Wash your hands or sanitise them upon entering/exiting the area.
- Sign in/register via the Tap In system or alternative system in place.
- Identify a seating area & log the desk number on 'Who Sat Where?' on the Udo app. For people who don't have access to the app inform your lecturer to use the alternative system.
- Always maintain social distancing.
- When the class finishes, please use the cleaning materials provided to clean down the area you have used & then wash or sanitise your hands.
- We welcome the wearing of face coverings in teaching areas where possible.

Thanks for your help and continued support.



IMPORTANT INFO!!

–BE AWARE – Not being paranoid but...

- There are cameras
- "They" listen to what we say (so that we can listen again later if we need to)
- Inform me if you object to recordings being made (and remind me every session!!)

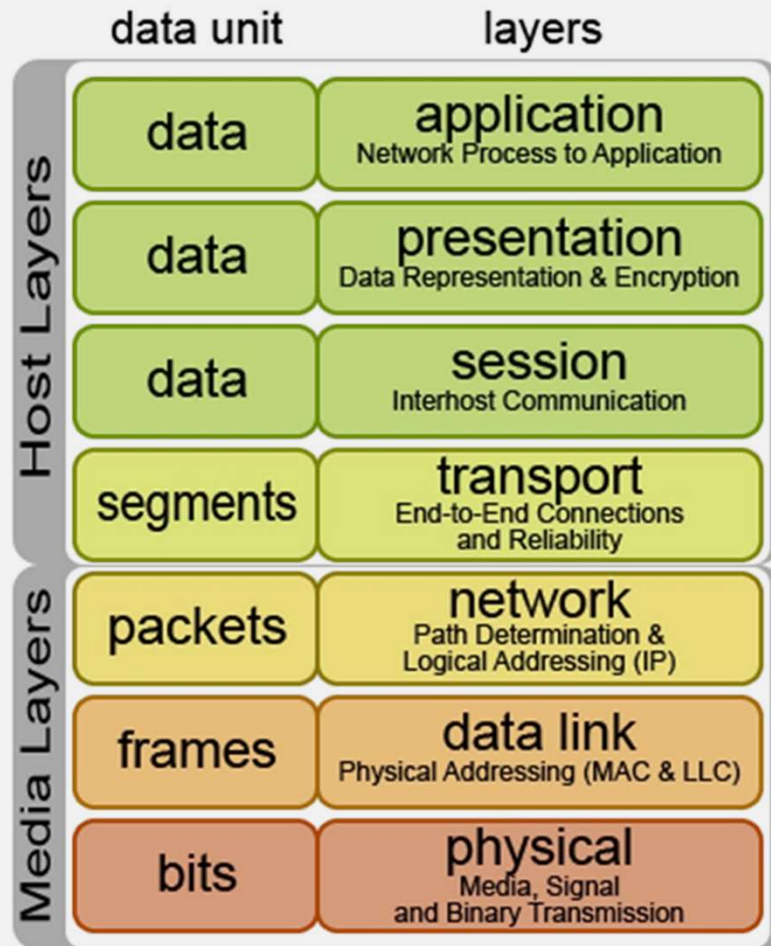
–TAP, TAP – only you and do stay for the lecture

- Attendance record is different to attending and learning

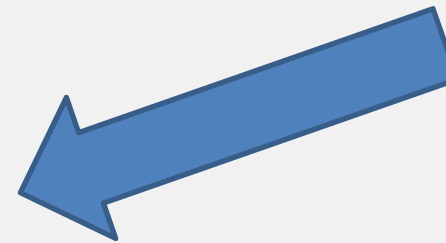
Week 5: Objectives

- Ethernet Protocol
- Address Resolution Protocol
- LAN Switches
- Summary

Layers 1 and 2 – the fun begins



- At the bottom of the model but important all the same



Layer 1 - Physical

- Where data, in the form of signals, moves from one device to another
- Could be pulses (levels) of electrical voltage
- Could be radio frequency signals
 - Might be affected by RFI (radio frequency interference)
 - Might be affected by EMI (electro magnetic interference)
- Could be pulses of light
 - Will not be affected by any radiated signals

Ethernet Standards (802.3)

- Ethernet standards
 - Standards make sure that we all do the same thing; this is good
- 802.3 standards define Layer 2 protocols and Layer 1 technologies

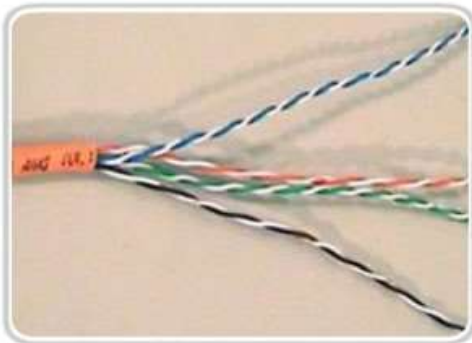
Ethernet Protocol (802.3)

- Ethernet –
 - Most widely used LAN technology
- Operates in the data link layer and the physical layer
- Supports data bandwidths of 10, 100, 1000, 10,000, 40,000, and 100,000 Mbps (100 Gbps)
- Suffix identifies subtypes e.g. 802.3a = 10base2 over coax, 802.3bz = 2.5base-T
 - not always speed!

Layer 1 - Physical

- Carry electrical signals over (copper) conductors
- Carry light through Fibre optic cables
- Carry radio frequency through space
- There are others, microwave, InfraRed, Sonic,
...

Copper Media



Unshielded Twisted Pair (UTP) cable

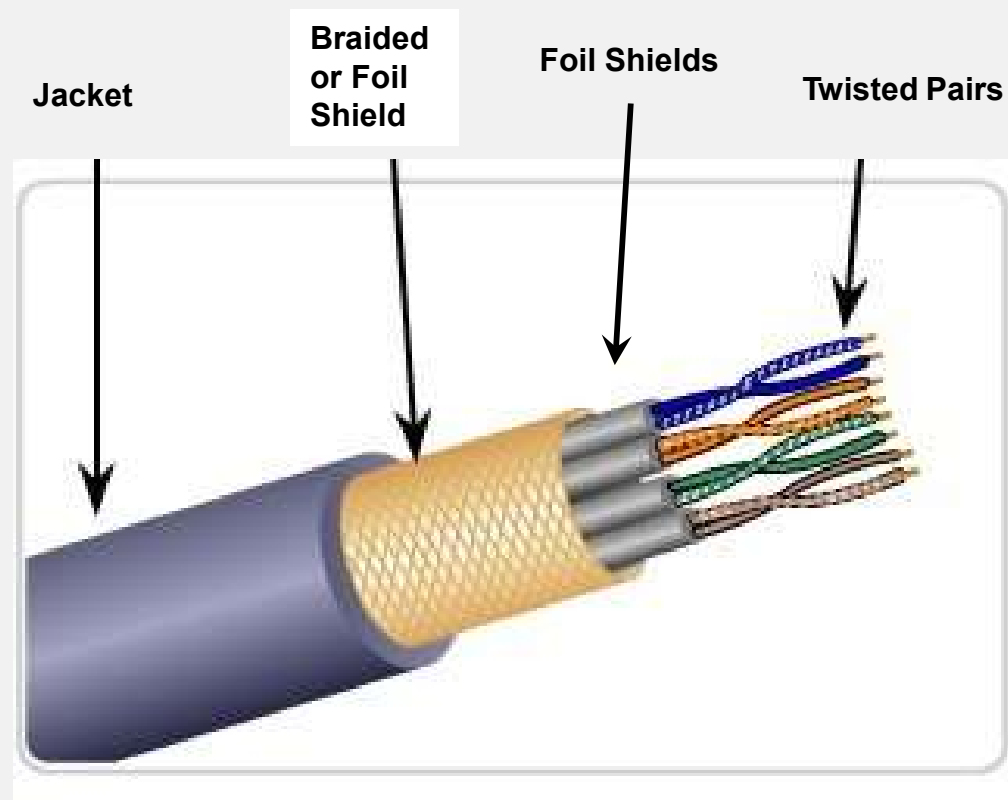


Shielded Twisted Pair (STP) cable



Coaxial cable

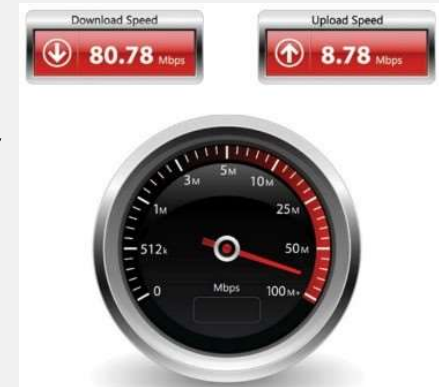
Shielded Twisted-Pair (STP) Cable



Copper “Categories”

Category	Shielding	Max Transmission Speed (at 100 meters)	Max Bandwidth
Cat 3	Unshielded	10 Mbps	16 MHz
Cat 5	Unshielded	10/100 Mbps	100 MHz
Cat 5e	Unshielded	1000 Mbps (1 Gbps)	100 MHz
Cat 6	Shielded or Unshielded	1000 Mbps (1 Gbps)	>250 MHz
Cat 6a	Shielded	10000 Mbps (10 Gbps)	500 MHz
Cat 7	Shielded	10000 Mbps (10 Gbps)	600 MHz

Bandwidth and Throughput



Bps = Bytes per second, bps = bits per second, **1Bps=8bps**

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	kbps	1 kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

bps = bits per second, Bps = Bytes per second, mBps = ???

Bandwidth and Throughput

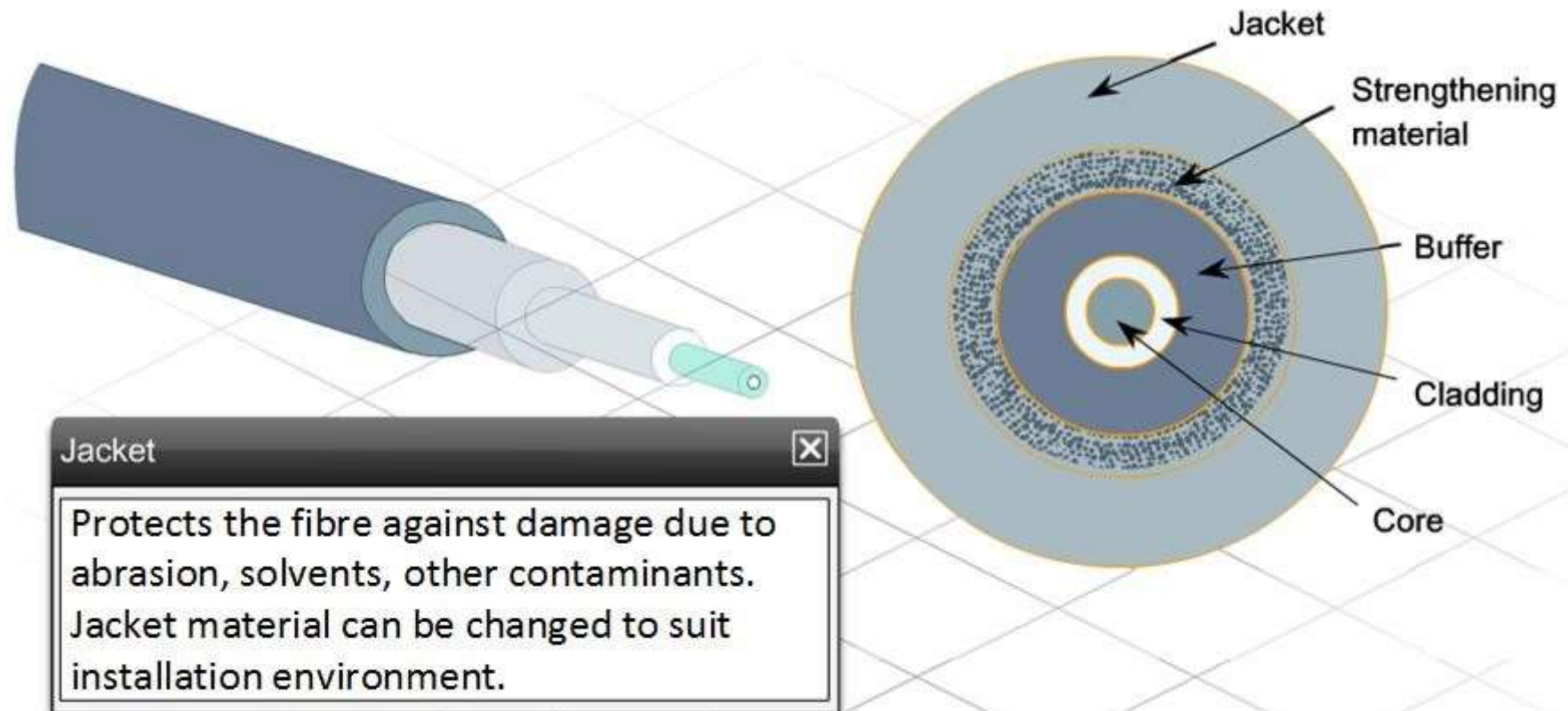


Bps = Bytes per second, bps = bits per second, **1Bps=8bps**

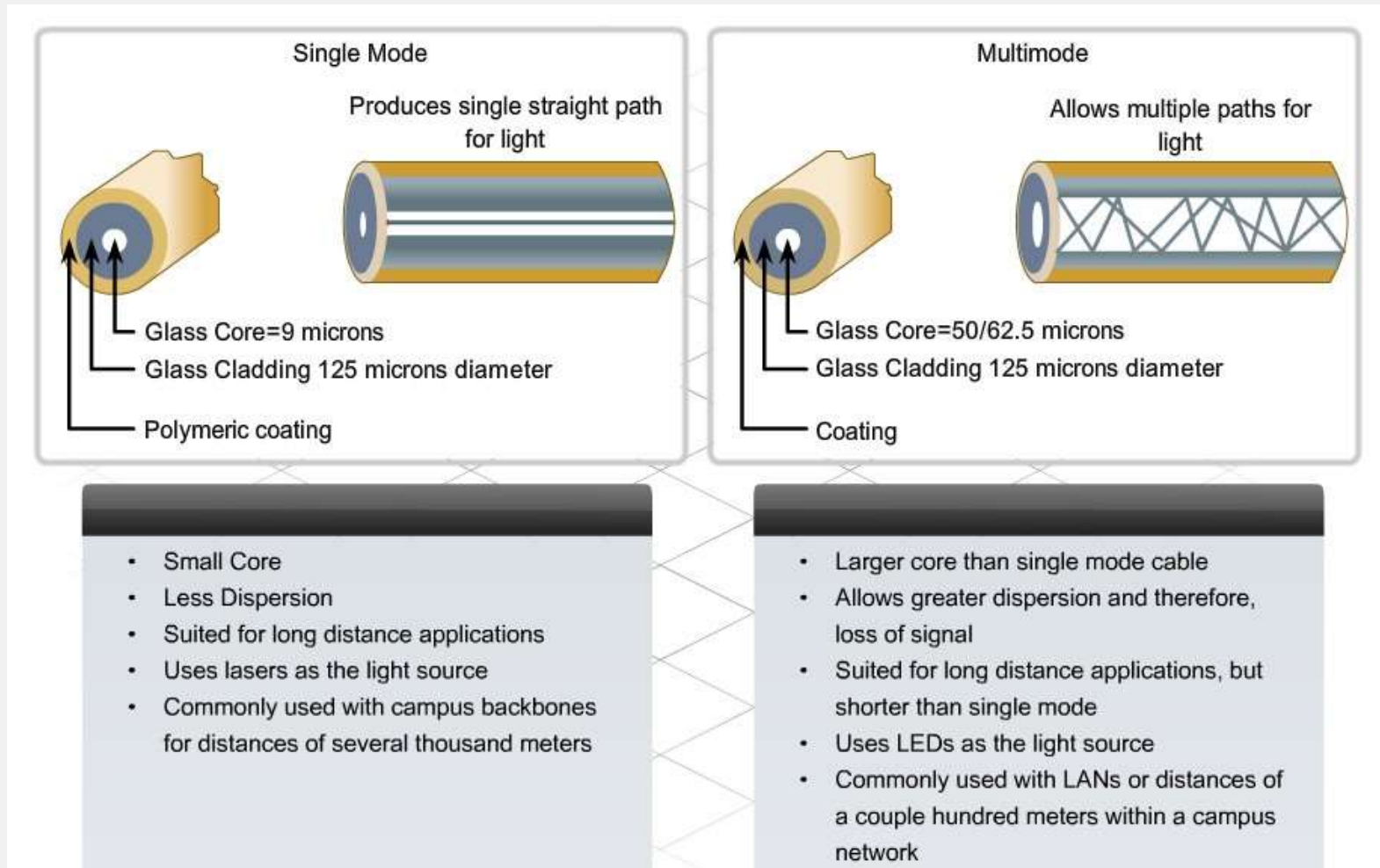
Unit of Bandwidth	Abbreviation	Equivalence
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Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

bps = bits per second, Bps = Bytes per second, mBps = milli-Bytes per second
(= 125 seconds per bit)

Fibre Optic Cabling



Types of Fibre Media



Fibre Optic Connectors (some)

- ST connector (aka "Shove Twist")
- SC connector (aka "Shove Click")

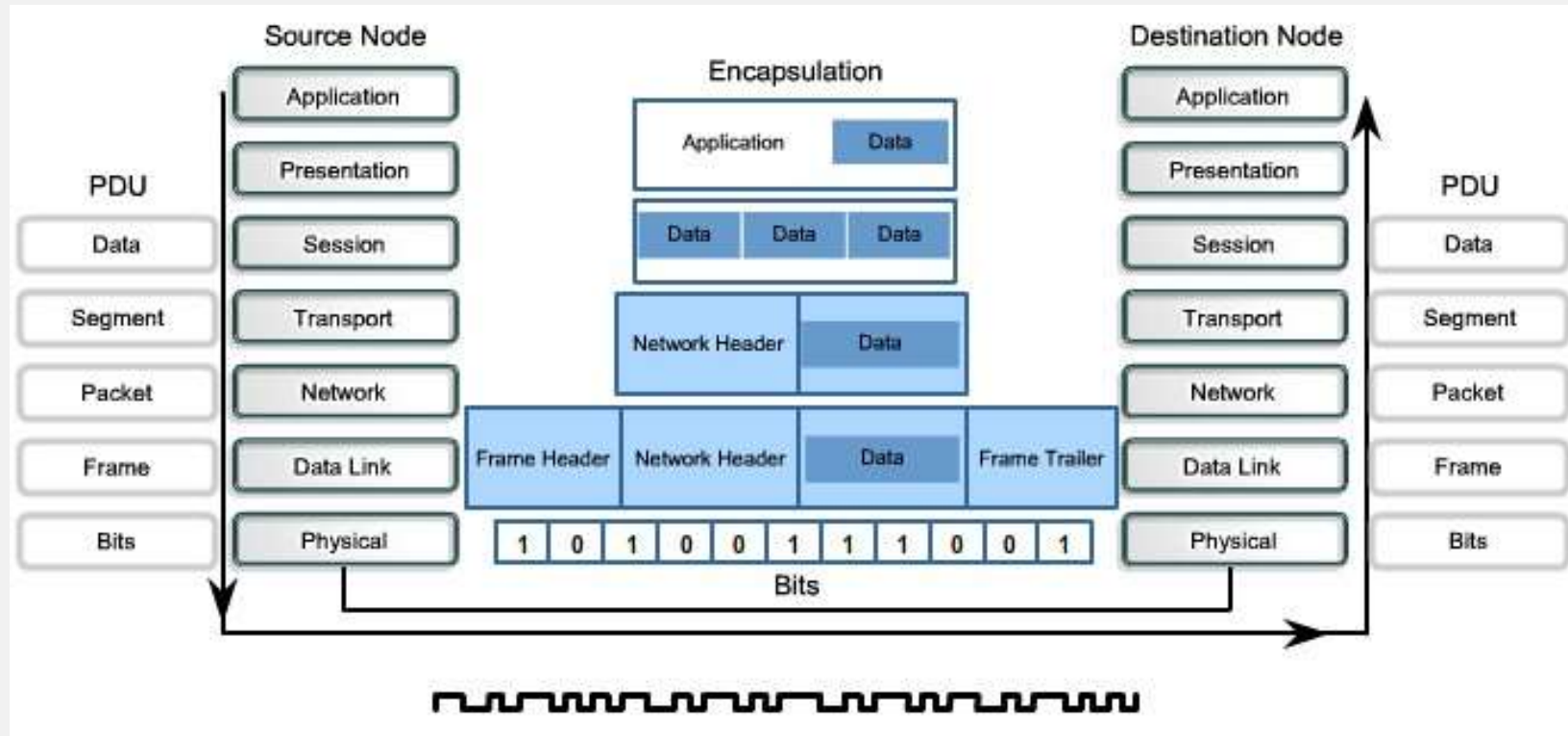


LC (little
connector)



MT - RJ (???)

The Physical Layer



Now we just need something to send...

LLC and MAC Sublayers

- LLC
 - Handles communication between upper and lower layers
 - Takes the network protocol data and adds control information to help deliver the packet to the destination
- MAC
 - Constitutes the lower sublayer of the data link layer
 - Implemented by hardware, typically in the computer network interface card (NIC)
 - Two primary responsibilities:
 - Data encapsulation
 - Media access control

MAC Sublayer

- Data encapsulation
 - Frame assembly and disassembly
 - MAC layer adds a header and trailer to the network layer PDU
- MAC Header functions:
 - Frame delimiting – group of bits indicate where frame begins and ends
 - Addressing – MAC address used to deliver to a destination node
 - Error detection – Uses a CRC check of the frame contents

Media Access Methods

- Carrier Sense Multiple Access / Collision Detection (**CSMA/CD**) process
 - Used to first detect if the media is carrying a signal
 - If no signal is detected, the device transmits its data
 - If two devices transmit at the same time - data collision
 - Wait a random amount of time before resending
- Original Ethernet used CSMA/CD

Media Access Methods

- CSMA/Collision Avoidance (**CSMA/CA**) media access method
 - Device examines the media for the presence of data signal - if the media is free, the device sends a notification across the media of its intent to use it
 - If no "objections" are received, the device sends the actual data
- Used by 802.11 wireless networking technologies

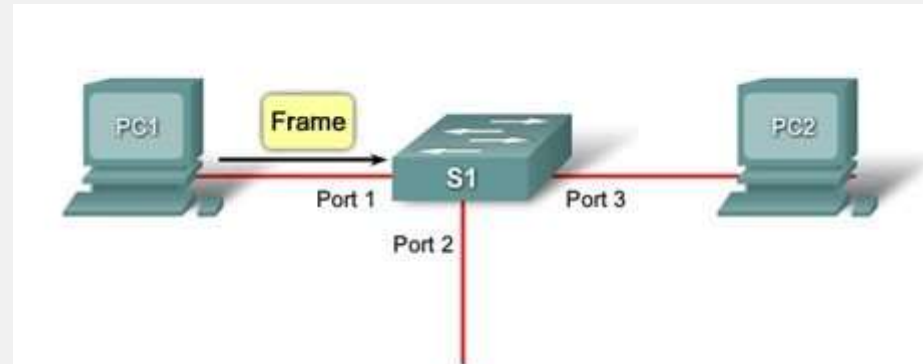
Frame Processing

- MAC addresses assigned to network cards of workstations, servers, printers, switches, and routers during manufacture
- Each NIC views information to see if the destination MAC address in the frame matches the device's physical MAC address
 - No match, the device discards the frame
 - Matches the destination MAC of the frame, the NIC passes the frame up the OSI layers, where further decapsulation takes place

Switch Port Fundamentals

- Layer 2 LAN switch
- Connects end devices to a central intermediate device on most Ethernet networks
- Performs switching and filtering based **only** on the MAC address
- Builds a MAC address table that it uses to make *forwarding* decisions
- Relies on routers to pass data between IP subnetworks

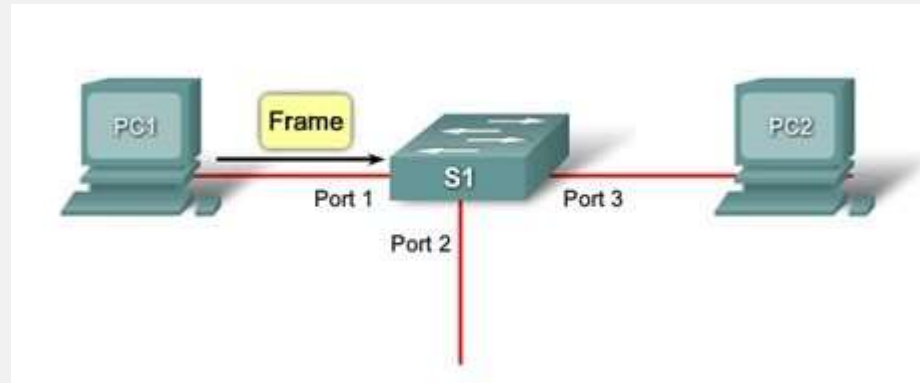
Building the Switch MAC Address Table



1. The switch receives a broadcast frame from PC 1 on Port 1.
2. The switch reads the source MAC address and records the ingress port, this is stored in the address table for use later...
3. Because the destination address is a broadcast, the switch **floods** the frame to all ports, except the port on which it received the frame.
4. The destination device (PC2) replies to the broadcast with a unicast frame addressed to PC 1.

- Continued...

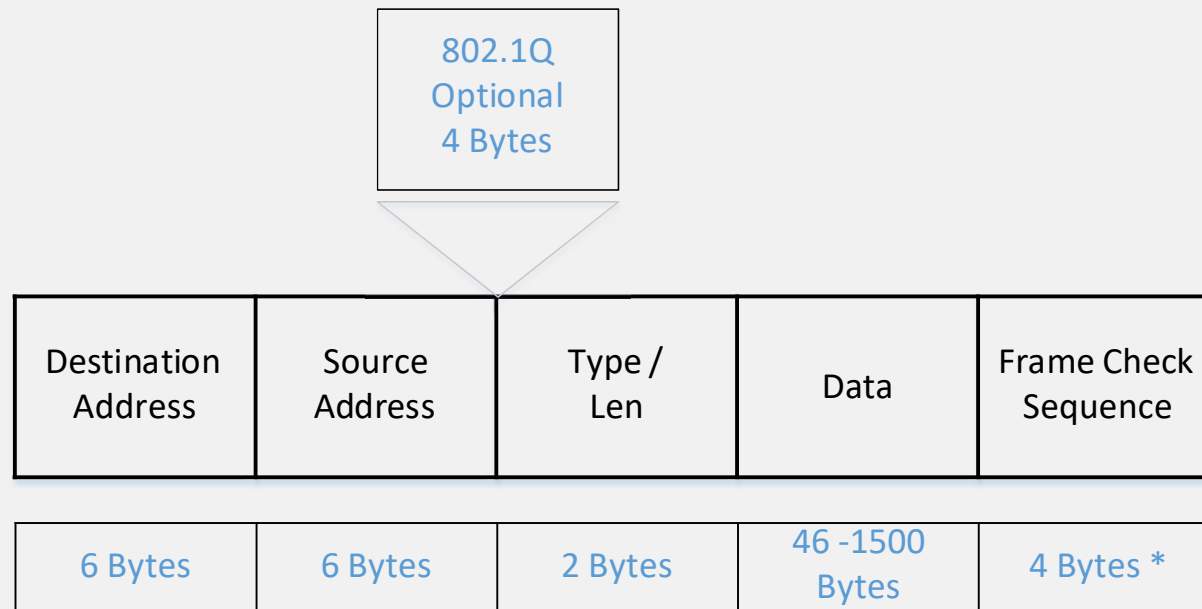
Building the Switch MAC Address Table



4. The switch reads the source MAC address of PC 2 from the reply and associates it with the ingress switch port number. This is recorded in the address table.
5. The two source addresses and ports are now held in the MAC address table.
6. The switch can now forward frames between source and destination devices directly. Flooding is not required and network traffic is reduced

Ethernet Frame Size

- Ethernet standards define the minimum frame size as 64 bytes and the maximum as 1518 bytes



Ethernet Frame Size

- The receiving device drops the frame if too large or small
- 802.1Q frames are 68 - 1522 bytes long (VLAN have extra 4 bytes, but skip this for now...)
- FCS is rarely passed through from the NIC – referred to as “offloading”
 - Let the NIC check the FCS, drop the packet if it is faulty, strip it if it was correct

MAC Address: Ethernet Identity

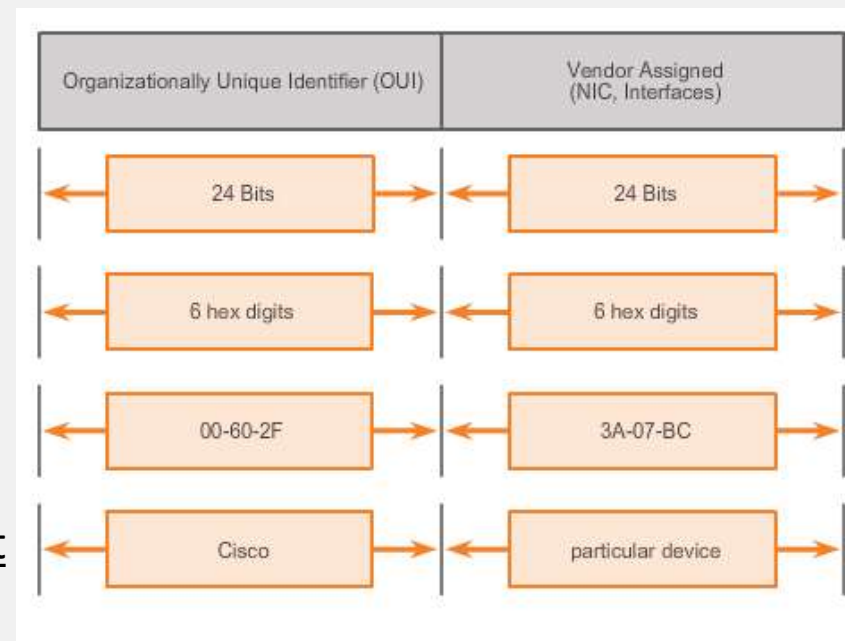
- Layer 2 Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits

IEEE specifies:

- First 3 bytes are vendor OUI (manufacturer code)
- Last 3 bytes, unique identity for part (serial number)

Current list - <http://standards-oui.ieee.org/oui.txt>

OUI = Organisationally Unique Identifier



(MAC Addresses and) Hexadecimal

Decimal and Binary equivalents of 0 to F Hexadecimal

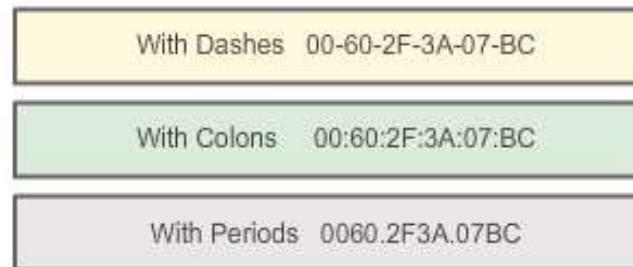
Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Selected Decimal, Binary and Hexadecimal equivalents

Decimal	Binary	Hexadecimal
0	0000 0000	00
1	0000 0001	01
2	0000 0010	02
3	0000 0011	03
4	0000 0100	04
5	0000 0101	05
6	0000 0110	06
7	0000 0111	07
8	0000 1000	08
10	0000 1010	0A
15	0000 1111	0F
16	0001 0000	10
32	0010 0000	20
64	0100 0000	40
128	1000 0000	80
192	1100 0000	C0
202	1100 1010	CA
240	1111 0000	F0
255	1111 1111	FF

MAC Address Representations

- Can appear in many forms depending on system:



```
C:\>ipconfig/all
```

```
Ethernet adapter Local Area Connection:
```

```
Connection-specific DNS Suffix  . : example.com
Description . . . . . : Intel(R) Gigabit Network Connection
Physical Address. . . . . : 00-21-CC-BA-44-C4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.1.67 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, November 26, 2012 12:14:48 PM
Lease Expires . . . . . : Saturday, December 01, 2012 12:15:02 AM
Default Gateway . . . . . : 192.168.1.254
DHCP Server . . . . . : 192.168.1.254
DNS Servers . . . . . : 192.168.1.254
```

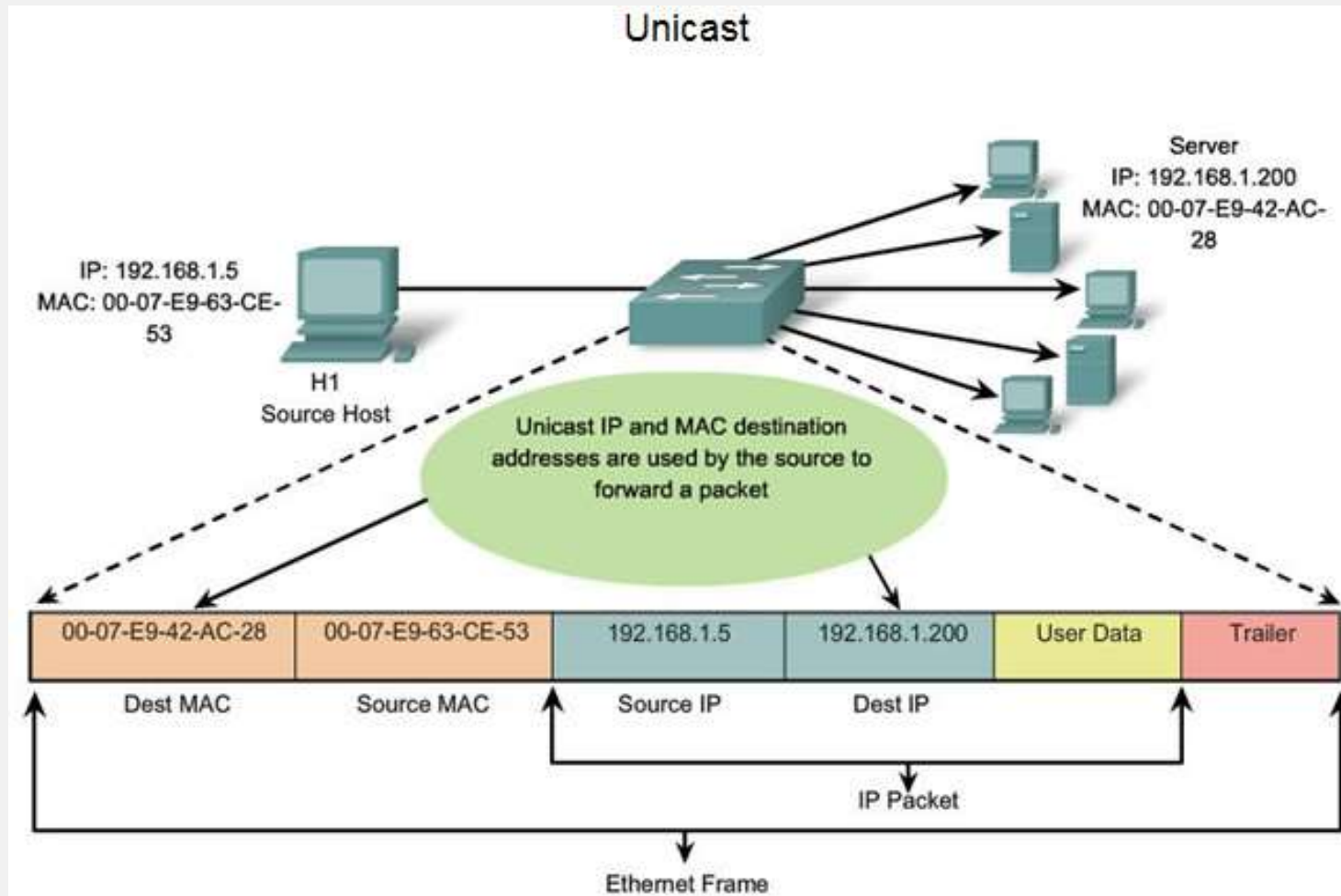
Unicasts, Multicasts and Broadcasts

- This is commonly associated with IP (Logical) addresses but it does apply to MAC (physical / hardware) as well
- The MAC table example had a destination IP address but not a destination MAC, broadcasting was necessary
- Broadcasts are typically sent to “all-ones” addresses (every bit of the address byte is set to 1; 6 “octets” x 8 bits = 48 ones)

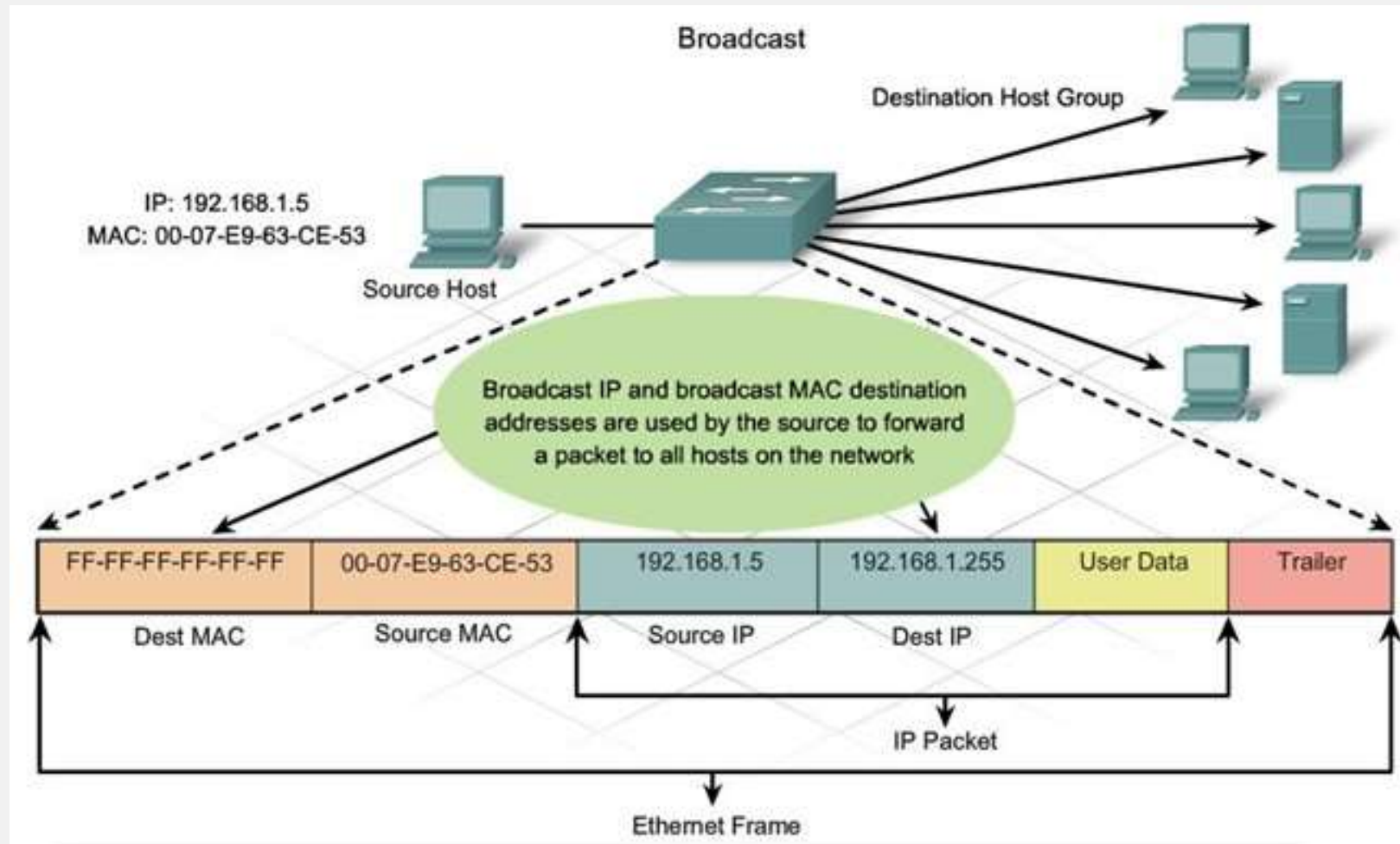
“All ones” addresses

- “all-ones” (and “all zeros”) addresses refer to binary representations of addresses
- You will need to be able to convert between Binary and Hexadecimal (and decimal)
- Binary “all ones” MAC address = 11111111
11111111:11111111 11111111:11111111
11111111
- Hex “all ones” MAC address = FFFF:FFFF:FFFF

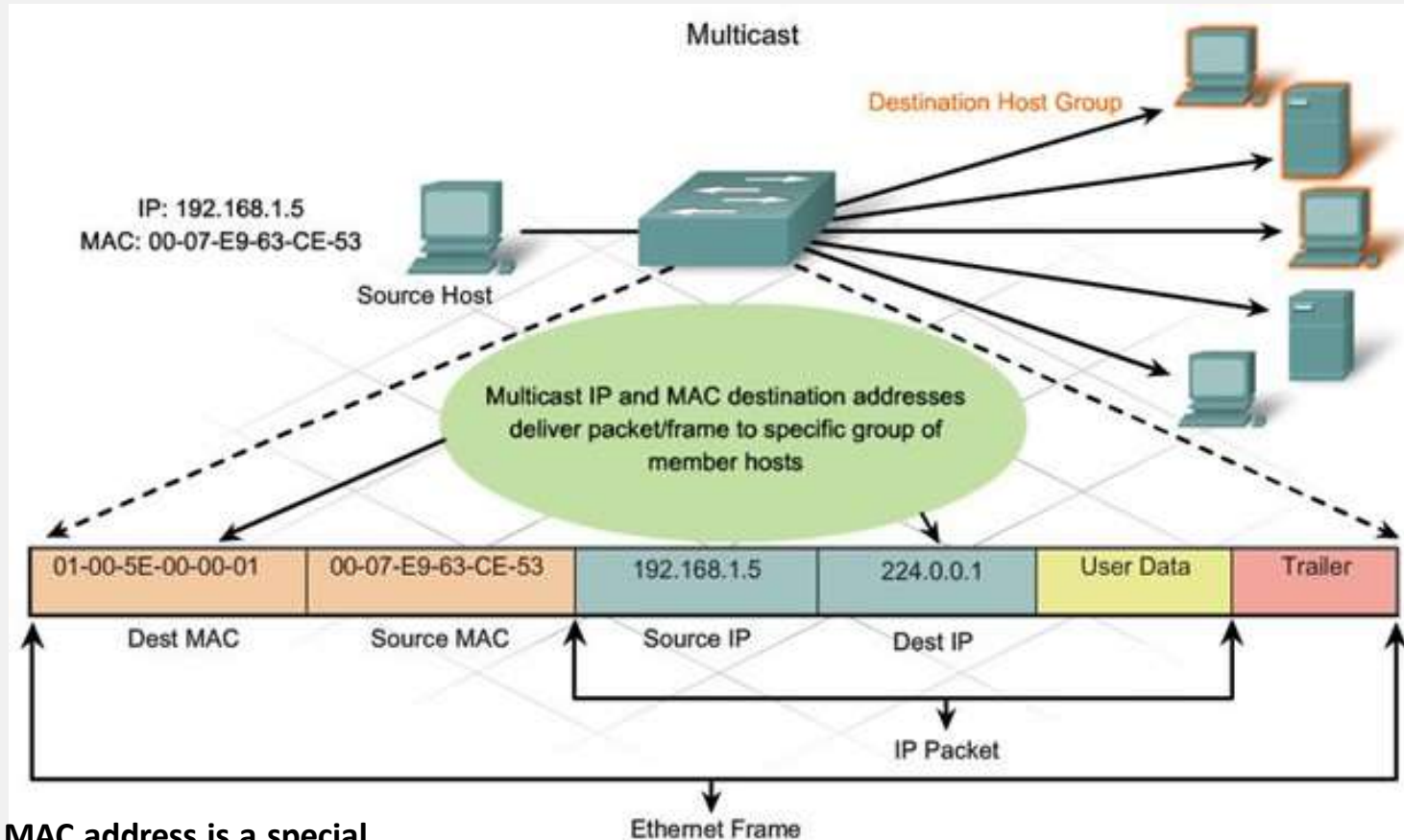
Unicast MAC Address



Broadcast MAC Address



Multicast MAC Address (advanced)



Multicast MAC address is a special value that begins with 01-00-5E in hexadecimal

Range of IPV4 multicast addresses is 224.0.0.0 to 239.255.255.255

MAC and IP

- MAC address

- This address does not change
- Similar to the name of a person
- Physical address (physically assigned to NIC)
- Assigned by manufacturer (under instruction from ieee)

- IP address

- Similar to the postal address of a person
- Based on where the host is actually located (now)
- Known as a logical address (logical view, not fixed forever)
- Assigned by a network administrator

Message delivery – End-to-End

- To deliver a message, a name and location is needed
- e.g. an email: userid (the name) @
 wherever.net (the location)
- e.g. a letter Mr B Strangler (a name)
 The Spooky House (a location)
 Shiver Lane

Message delivery – End-to-End

- Network messages follow the same rules
- Need a hardware address (fixed name) and a logical IP address (**Both**)

End-to-End Connectivity, MAC, and IP

Destination MAC Address BB:BB:BB:BB:BB:BB	Source MAC Address AA:AA:AA:AA:AA:AA	Source IP Address 10.0.0.1	Destination IP Address 192.168.1.5	Data	Trailer
--	---	-------------------------------	---------------------------------------	------	---------

A switch examines MAC addresses.

Destination MAC Address BB:BB:BB:BB:BB:BB	Source MAC Address AA:AA:AA:AA:AA:AA	Source IP Address 10.0.0.1	Destination IP Address 192.168.1.5	Data	Trailer
--	---	-------------------------------	---------------------------------------	------	---------

A router examines IP addresses

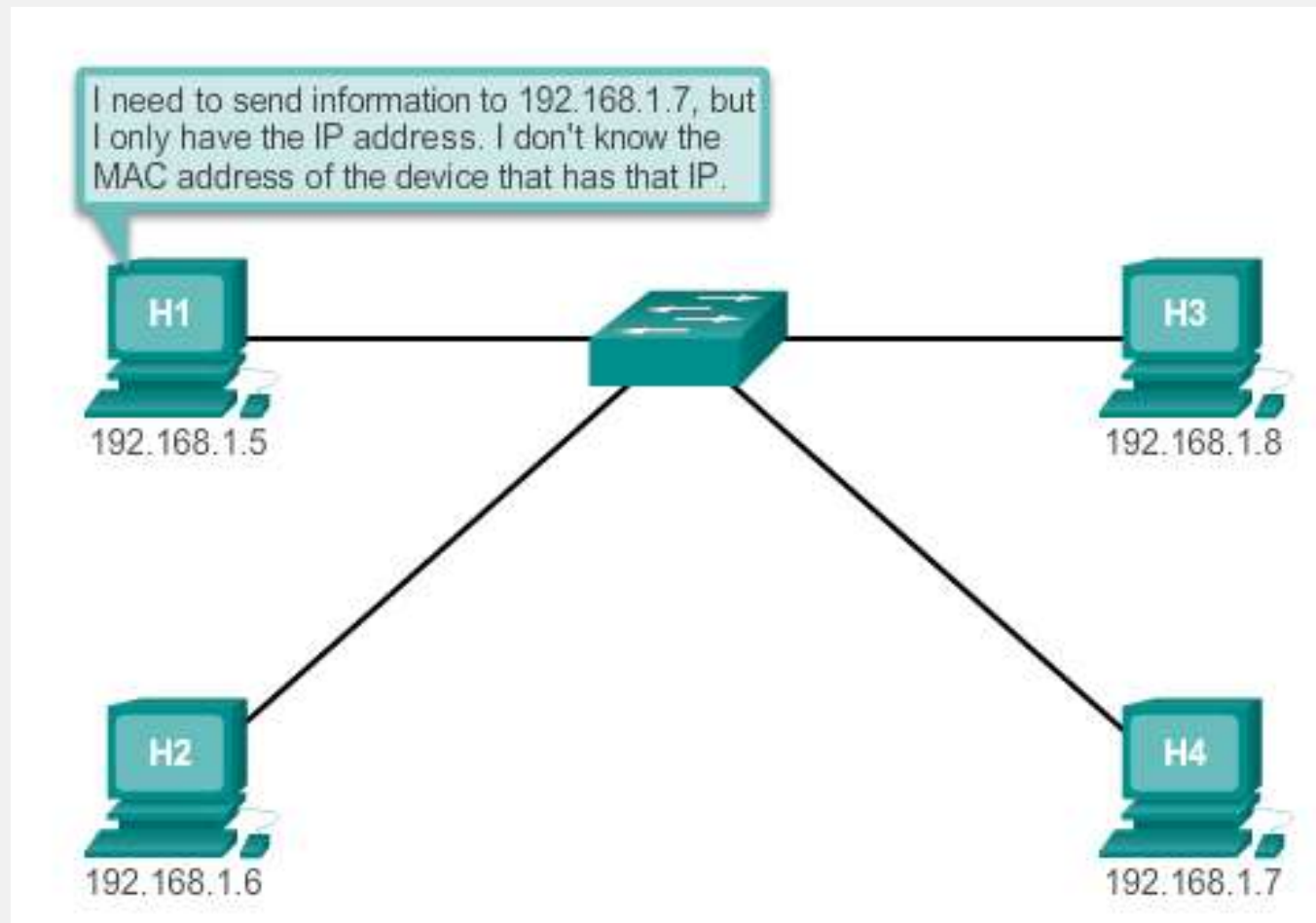
Introduction to ARP

- Easy to send a message to an IP address, we can set that according to our cloud
- Not easy to know the MAC address, globally unique, set by the manufacture, 1 digit different could be in another continent
- Need a method to resolve the hardware/physical/MAC address to an IP/logical address
- Luckily ARP, the Address Resolution Protocol came along...

Introduction to ARP

- Used when a sending “node” needs to find the MAC address of a destination “node” on the LAN
- The ARP protocol does two things:
 - Resolves IPv4 addresses to MAC addresses
 - Maintains a table of mappings (IP to MAC relationships)

Introduction to ARP



ARP Functions/Operation

- ARP Table
- As a node receives frames from the media, it records the source IP and MAC address as a mapping in the ARP table
- ARP request
- Layer 2 broadcast to all devices on the Ethernet LAN (Contains the specific IP address but an “all ones” MAC)
- The node that matches the IP address in the broadcast will reply
- If no device responds, packet is dropped (the device does not exist)

ARP Role in Remote Communication

- If the destination IPv4 host is on the local network, the frame will use the MAC address of this device as the destination MAC address
- If the destination IPv4 host is not on the local network, the source uses the ARP process to determine a MAC address for the router interface serving as the gateway
- In the event that the gateway entry is not in the table, an ARP request is used to retrieve the MAC address associated with the IP address of the router interface

ARP Tables on Networking Devices

```
Router#show ip arp
```

	Protocol	Address	Age (min)	Hardware Addr	Type	Interface
	Internet	172.16.233.229	-	0000.0c59.f892	ARPA	Ethernet0/0
	Internet	172.16.233.218	-	0000.0c07.ac00	ARPA	Ethernet0/0
	Internet	172.16.168.11	-	0000.0c63.1300	ARPA	Ethernet0/0
	Internet	172.16.168.254	9	0000.0c36.6965	ARPA	Ethernet0/0

```
C:\>arp -a
```

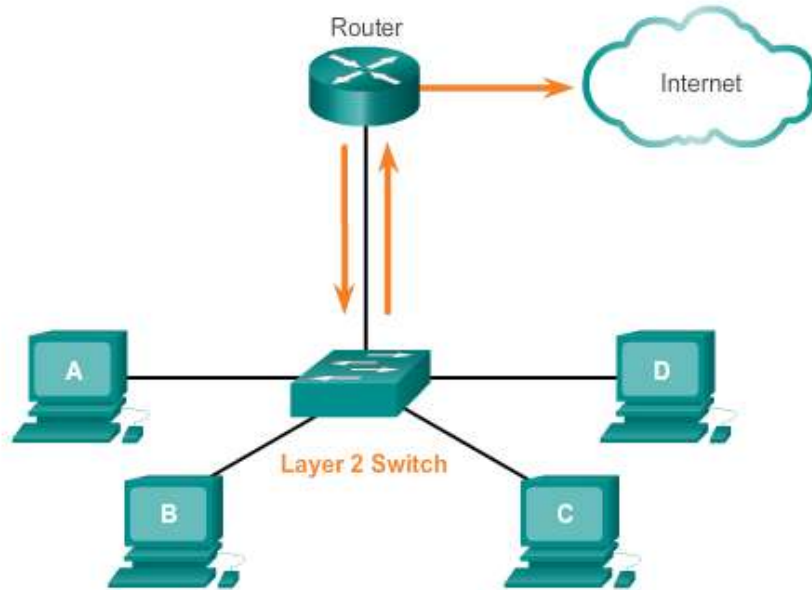
```
Interface: 192.168.1.67 --- 0xa
```

Internet Address	Physical Address	Type
192.168.1.254	64-0f-29-0d-36-91	dynamic
192.168.1.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

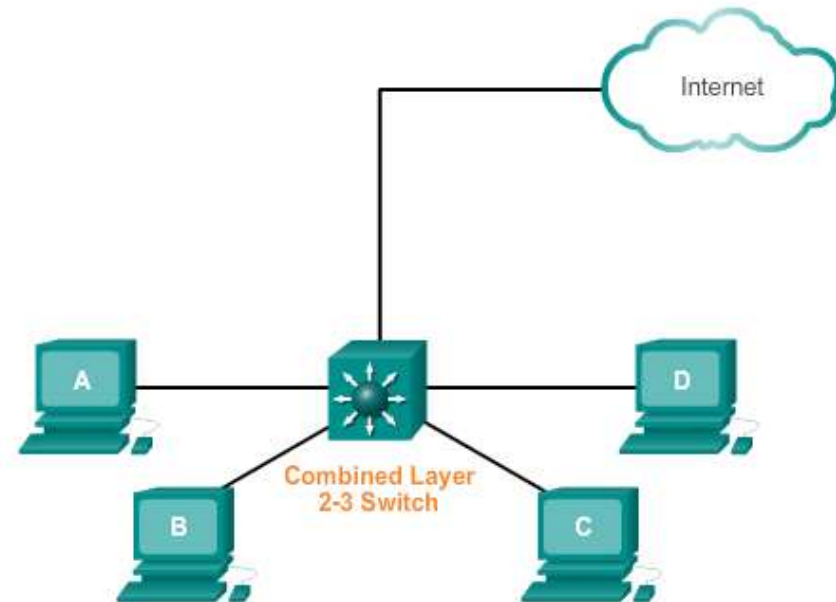
Layer 2 verses Layer 3 Switching

– a problem for another day

Layer 2 Switching



Layer 3 Switching



Summary

- Ethernet standards define both the Layer 2 protocols and the Layer 1 technologies.
- The Ethernet frame structure adds headers and trailers to encapsulate the message being sent
- The Ethernet frame provides MAC addressing and error checking
- Each node on an IP network has both a MAC address and an IP address

Summary

- Ethernet uses ARP to determine the MAC addresses of destinations and map them against known IP addresses.
- A Layer 2 switch builds a MAC address table that it uses to make forwarding decisions
- Layer 3 switches do something else

